



LABORATORY ACTIVITY NO. 1.5.6

Using a Search Engine

This lab activity, "Using a Search Engine," is designed to enhance the students' proficiency in utilizing search engines as a crucial troubleshooting tool within a **Windows 11** environment. The scenario involves addressing a common issue where a laptop cursor jumps erratically during typing. You will learn to effectively use a search engine (like Google) to research technical problems, analyze search results to identify relevant information (including forum discussions and driver downloads), and understand the process of locating specific drivers (e.g., touchpad drivers) on a manufacturer's support website. The lab emphasizes critical thinking and effective search strategies for diagnosing and resolving computer issues.

I. OBJECTIVES

Upon completion of this laboratory exercise, students are expected to and meet the following intended learning outcomes:

Activity Objectives	Intended Learning Outcomes		
	ILO1	ILO2	ILO3
1) effectively utilize a search engine to research and troubleshoot common computer hardware issues.	•	•	
2) configure web browser settings, including changing the default search provider.	•	•	
3) analyze search results to identify relevant technical information, distinguish between different types of sources, and locate official support and driver download pages.	•	•	•

Where:

- ILO1 – Exhibit thorough demonstration of the concept
- ILO2 – Being able to perform the activity independently
- ILO3 – Design and evaluate emerging IT solutions

II. DISCUSSION

- **Support and Troubleshooting:**
 - Troubleshooting is defined as a process of problem-solving.



- Problems consist of causes, symptoms, and consequences.
- The importance of understanding priorities and business needs in troubleshooting is highlighted.
- The concept of "first response" often involves simply rebooting the system.
- Effective troubleshooting relies on good documentation, including user manuals, support websites, and knowledge bases.
- It emphasizes identifying potential causes through organized data gathering.
- A six-step basic troubleshooting procedure is outlined: identify the problem, establish a theory of probable cause, test the theory, establish a plan of action, verify full system functionality, and document findings.

- **Customer Service:**

- Good communication skills are essential in technical support roles.
- Active listening and setting clear expectations (Service Level Agreements - SLAs) are important.
- Conflict resolution techniques involve showing empathy, managing user frustration, and avoiding technical jargon.
- Respecting user privacy is a critical aspect of customer service.

- **Technical Support:**

- When contacting technical support, specific information should be provided, including contact details, details of the software or device, version and model numbers, purchase date, and a clear description of the problem.
- Utilizing technical community groups and forums for support is also discussed.

- **Using a Search Engine:**

- A search engine is described as a tool for locating web pages, capable of searching the entire web or specific sites.

III. MATERIALS

A Windows 11 installed computer system.

IV. PROCEDURE

For this activity, you will execute the provided step-by-step instructions. As you complete each step, capture a screenshot of the resulting output or completed action that directly demonstrates successful execution of that instruction. Paste each screenshot into the designated area within your lab document.

Important Considerations:



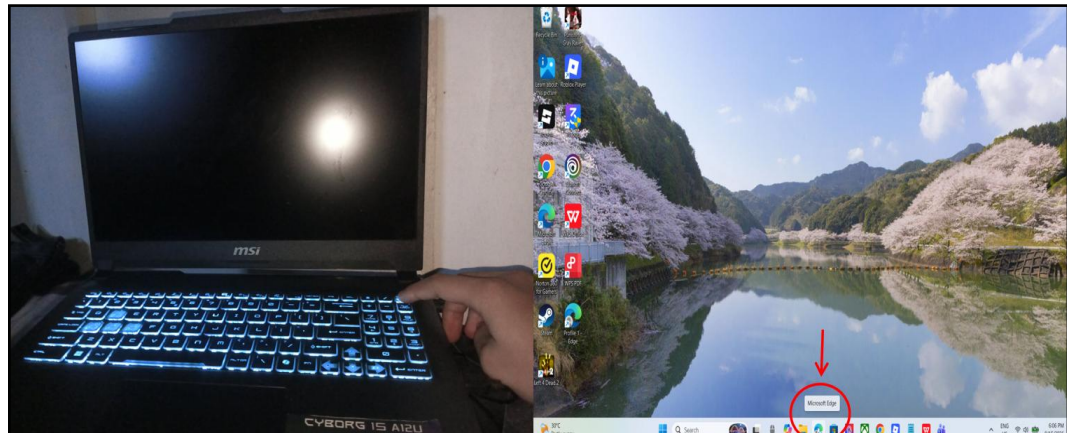
- **Operating System:** The core functionality of web browsers and search engines remains consistent in both Windows 10 and 11. However, the visual interface of **Microsoft Edge (the default browser in Windows 11)**, the Settings app, and some navigation paths might appear slightly different.
- **Internet Connection:** A stable internet connection is essential for this lab, as it relies heavily on accessing online search engines and manufacturer websites.
- **Browser Choice:** The lab specifically uses Microsoft Edge. Users should be comfortable with Edge's user interface and features, although the principles learned are applicable to any modern web browser.
- **Search Engine Settings:** The lab involves potentially changing the default search provider within your browser. In Windows 11's Edge, this process is done through the browser settings.
- **Troubleshooting Mindset:** Approach the search process systematically. Refine your search queries and critically evaluate the credibility and relevance of search results, distinguishing between advertisements, forum discussions, and official support pages.
- **Driver Downloads and System Changes:** The lab guides you to the point of identifying a specific driver but **explicitly states not to proceed with downloading and installing it**. In a real troubleshooting scenario, always exercise caution when installing drivers. It is recommended to back up your system and verify the authenticity of the driver source before proceeding with any installation. This lab focuses solely on the *search and identification* aspects of the process.
- **Forum Etiquette:** The lab briefly touches upon navigating online forums. In a real troubleshooting context, providing specific details like the exact model number of your device when posting on forums is crucial for receiving effective support.

Members:

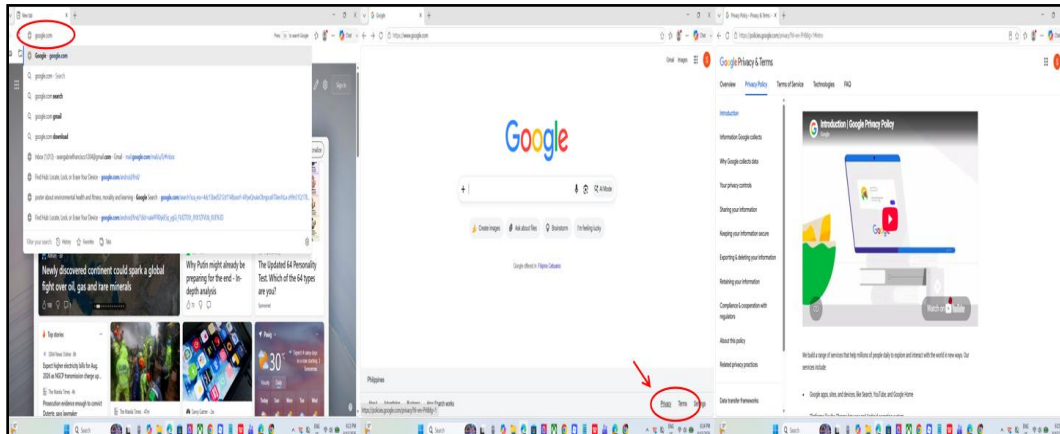
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Steps:

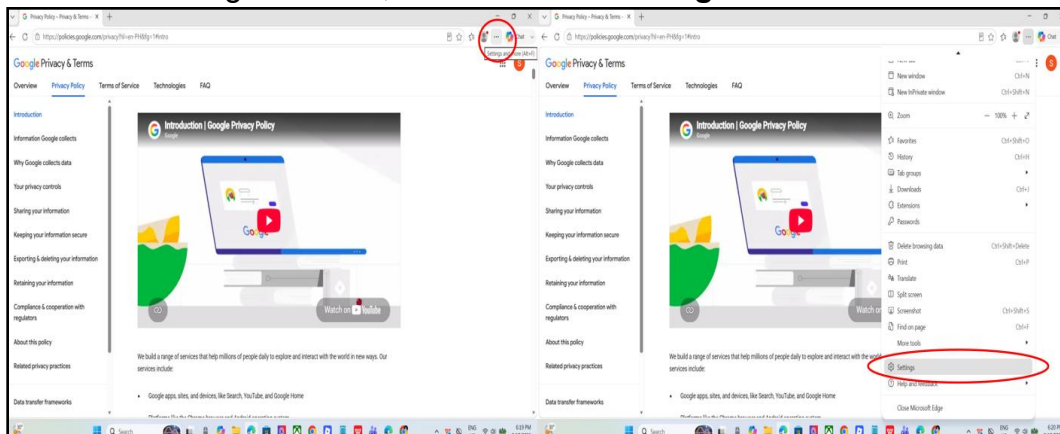
1. If necessary, start your computer and sign in. Then, open the **Microsoft Edge** browser from the taskbar (it's usually pinned in the center or left side).
 - *Note:* When you buy a computer from an OEM, you might want to change the default search provider. For Edge, the engine must support OpenSearch (opensearch.org), and you must have visited the search engine's home page at least once.



2. In the Edge browser, open google.com (you will likely be redirected to the Google home page for your country). If prompted, complete any wizards explaining Google's privacy and data collection policies.

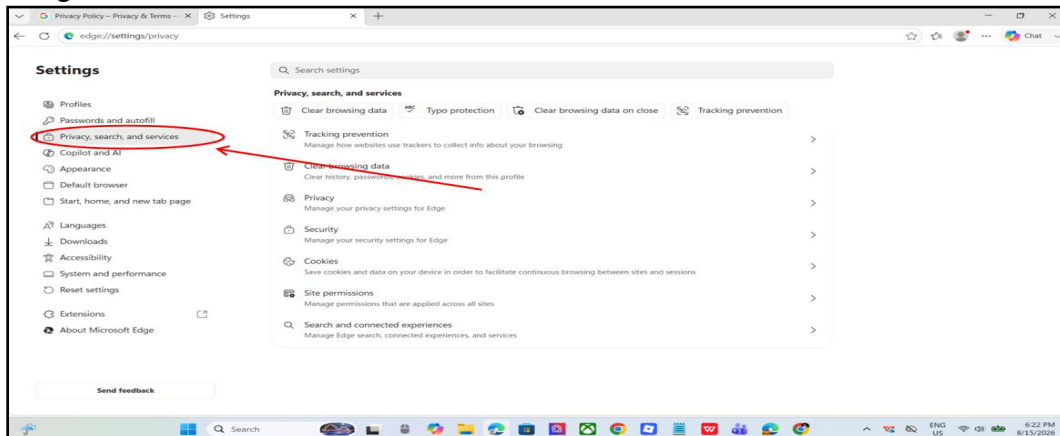


3. Click the "... (three dots)" icon for "Settings and more" located in the top-right corner of the Edge browser, and then click **Settings**.

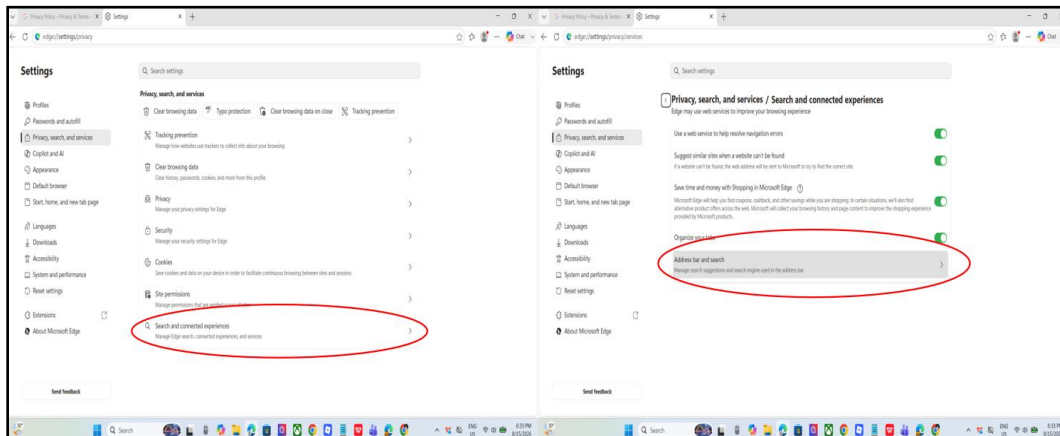




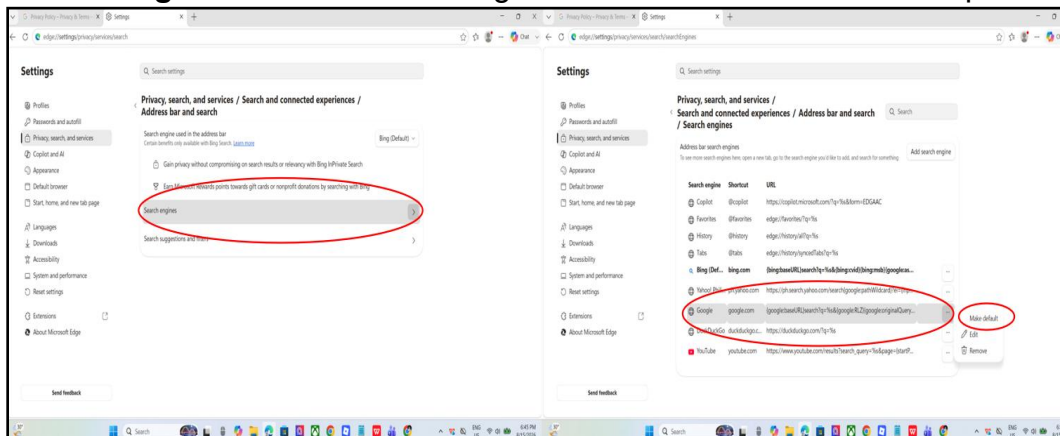
4. In the Settings pane, click **Privacy, search, and services** in the left-hand navigation.



5. Scroll down to the "Services" section and click **Address bar and search**.

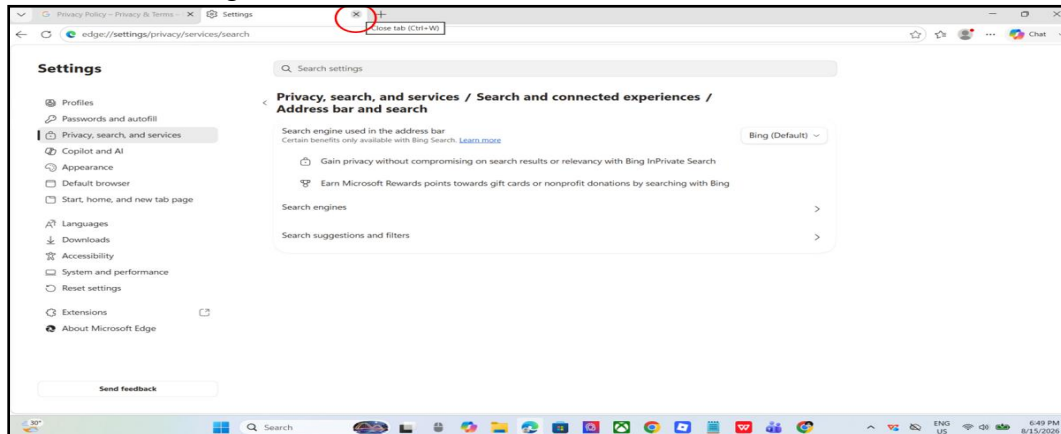


6. If you want to change the default search provider (you do not have to for this lab), select **Google** from the "Search engine used in the address bar" dropdown menu.

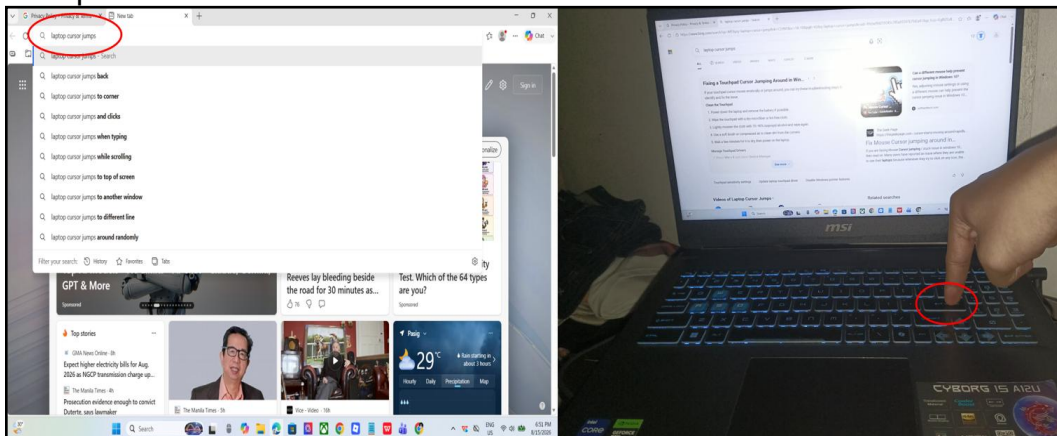




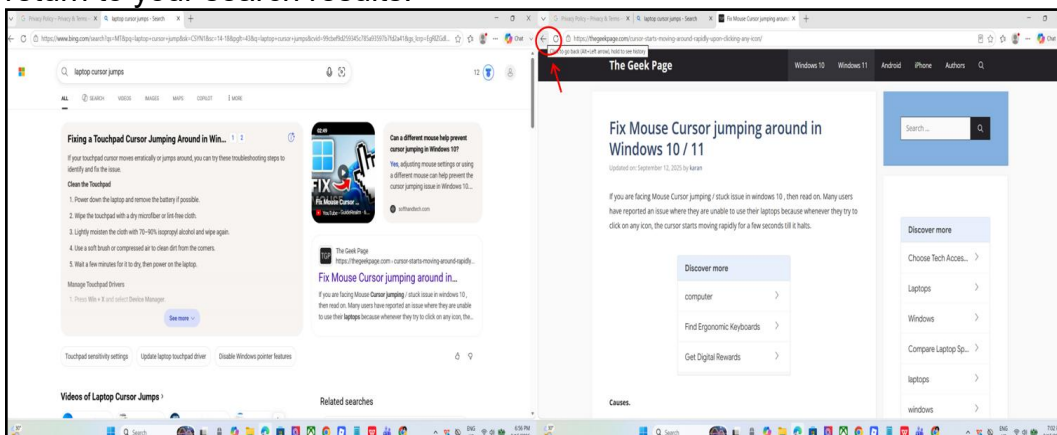
7. Close the Settings tab.



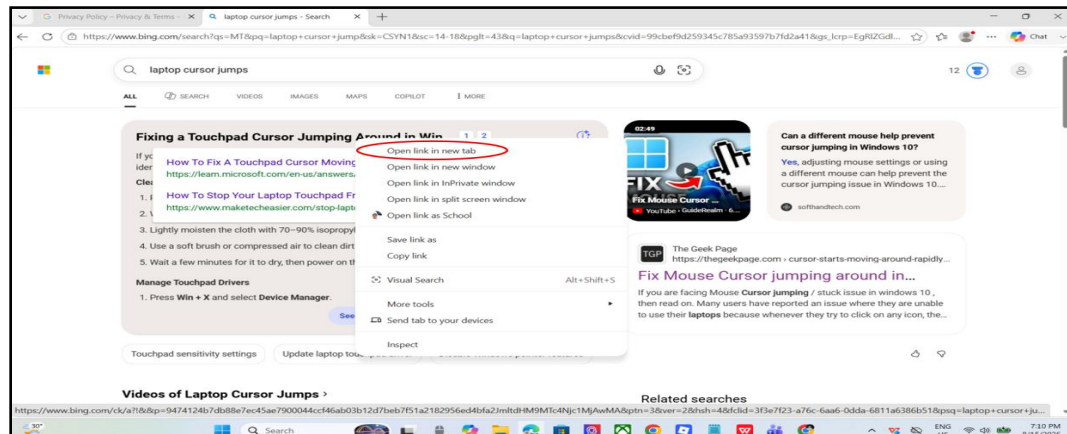
8. Click in the address bar of your Edge browser, type laptop cursor jumps, and then press ENTER.



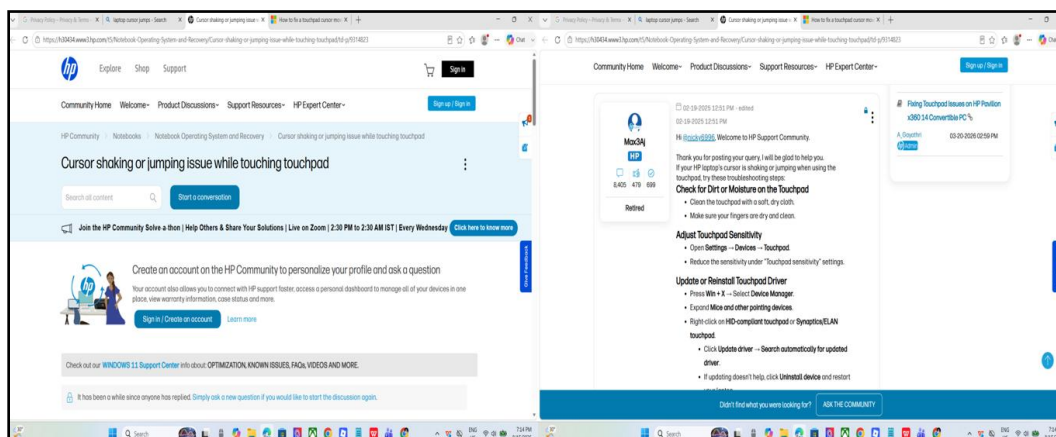
9. Browse through the search results. You should find a mix of ad-supported sites explaining the issue and forums with multiple posts relating to similar issues. Take some time to explore some of these, using the **Back** button in Edge to return to your search results.



- Alternatively, right-click a search result and select "Open link in new tab" to keep your search results open in a separate tab.

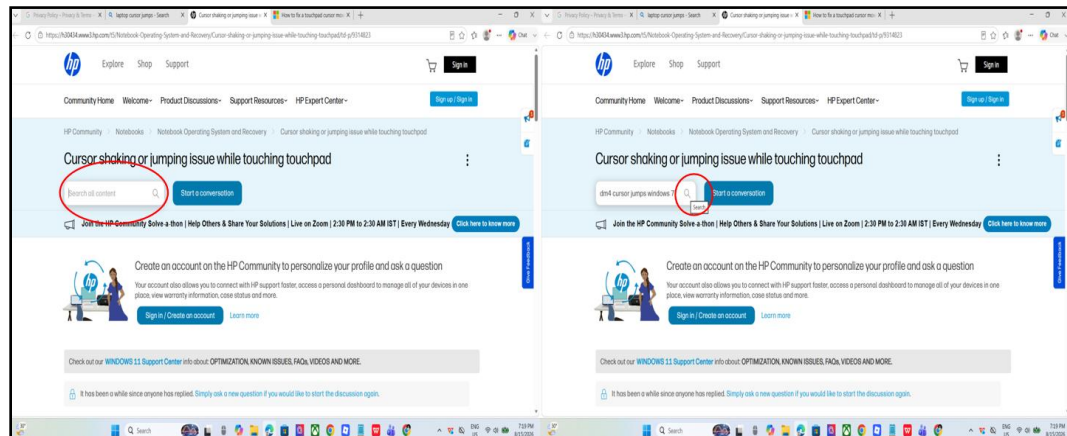


10. Pick one of the HP forum pages (e.g., "h30434.www3.hp.com") from the search results, and read through some of the posts to understand common solutions or discussions.

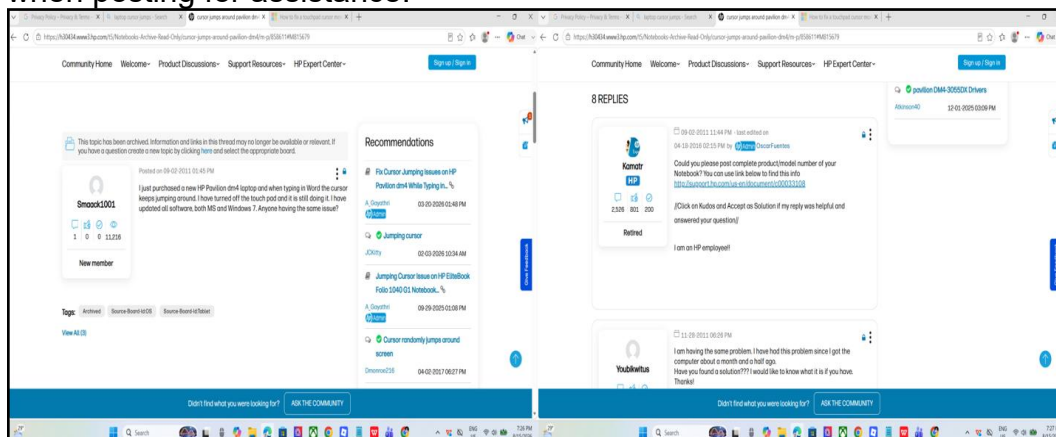


11. Within the HP forum, locate the forum's search box and type dm4 cursor jumps "windows 7". Change the list box option (if available) to "Entire Forum" and click the search button (often labeled "Go" or a magnifying glass icon).

- **Note:** While you are using Windows 11, searching for solutions related to older Windows versions like Windows 7 can still yield relevant troubleshooting steps, especially for hardware-specific issues.



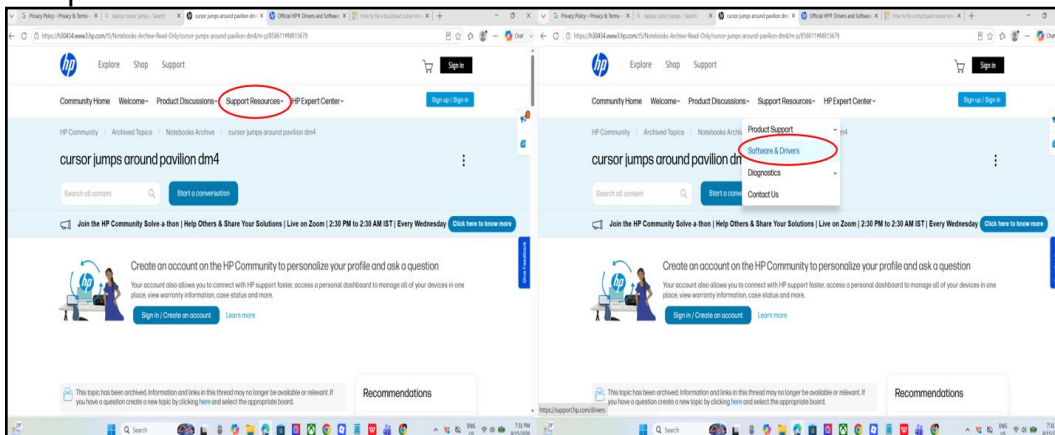
12. Read some of the posts and their replies within the forum. Pay attention to the frequent request for users to specify the exact model number of their device when posting for assistance.



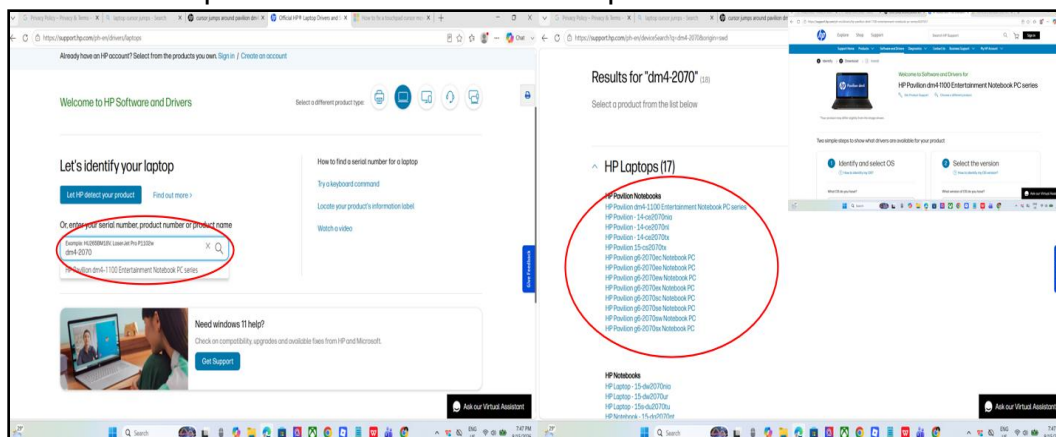
You decide to locate the latest touchpad driver version, as the configuration settings for palm rest do not seem to be present in the version you currently have.



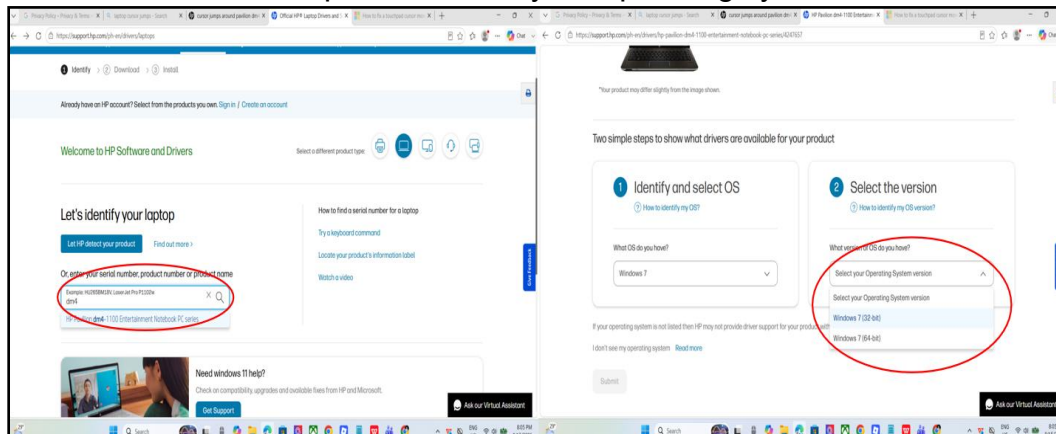
13. On the HP forum page, try to locate a **Support** or **Drivers** menu link and select an option like "Download drivers" or "Software and Drivers."



14. In the product search box on the manufacturer's support site, type dm4-2070 (this is a placeholder for a specific laptop model) and then click the search or "Go" button. Open the link for the matched product.

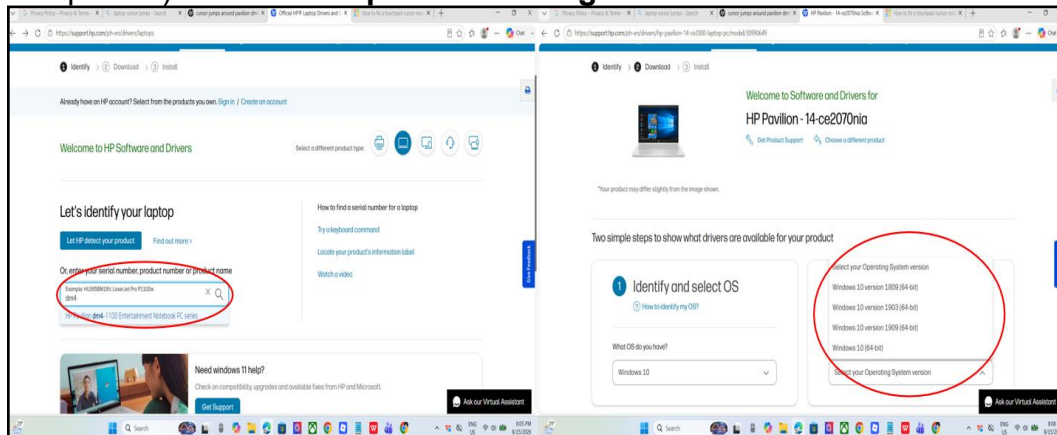


15. Note if there's an option to use a software tool to identify missing drivers. For this lab, proceed by locating the drivers manually, typically by clicking a "Go" or "Find drivers" link under an option to select your operating system.

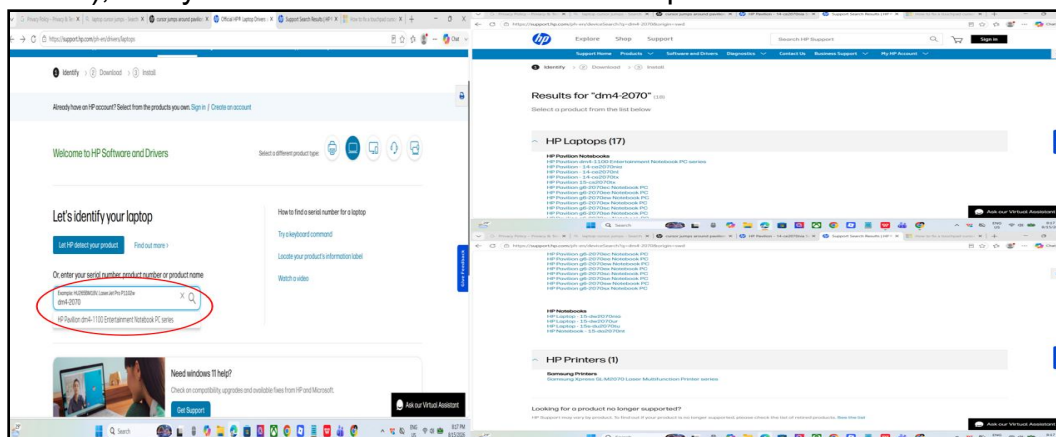




16. Choose the **OS option as Windows 11** (or the closest available option like Windows 10 if Windows 11 isn't explicitly listed, as many drivers are cross-compatible). Then click **Update** or **Change**.



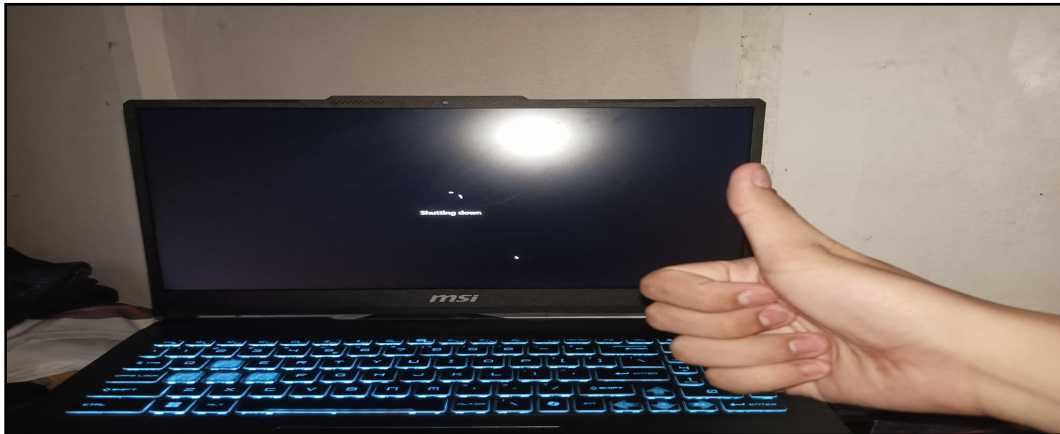
17. Expand the category like "Driver-Keyboards, Mouse and Input Devices" (or similar), and you should see a link for the touchpad driver.



We will not continue from here. In a real-world scenario, the next steps would be to ensure the user's computer is backed up, then download and install the identified driver. It can be difficult to immediately confirm if a problem like an erratic cursor has been solved, so you would typically encourage the user to report back after a day or two to confirm whether the cursor issue is still present.



18. Shut down your computer.



V. REFLECTION:

Upon successful completion of "Lab 1.5.6: Using a Search Engine," write a comprehensive reflection (between 150-250 words) addressing the following points:

- **Learning Outcomes:** What new knowledge or skills did you acquire or reinforce regarding the use of a search engine?
- **Challenges and Solutions:** Describe any difficulties or unexpected issues you encountered during the lab. How did you overcome them, or what steps would you take to resolve them in the future?
- **Real-world Relevance:** How do the skills and understanding gained in this lab relate to real-world scenarios, either in your personal life or a professional IT context?
- **Critical Thinking:** If you had to use an internet image for any step, explain why you couldn't perform the action physically and how finding the image contributed to your understanding.

We reinforced our understanding of efficient search engine use, learning advanced operators like quotation marks for exact phrases, the minus sign to exclude irrelevant terms, and site-specific searches to narrow results to trusted domains. We also practiced evaluating source credibility by checking URLs, publishers, and publish dates before trusting information found online.

The main challenge we encountered was unfamiliarity with the HP website's layout, which made it harder to locate specific pages and information during the lab. We resolved this by taking time to explore the site's navigation and menus until we found what we needed. Going forward, we'd familiarize ourselves with a site's structure early on before searching for specific content.

These skills are directly relevant to real-world IT work, where troubleshooting often depends on quickly finding accurate technical documentation, error code explanations, or vendor



support pages. Being able to navigate unfamiliar websites and filter noise from search results efficiently is a practical time-saver both in coursework and on the job.

For every step in this lab, we were able to perform the actions physically and captured our own original screenshots rather than relying on internet images. This allowed us to document our actual process accurately and ensured our reflection reflects genuine hands-on experience rather than secondhand references.